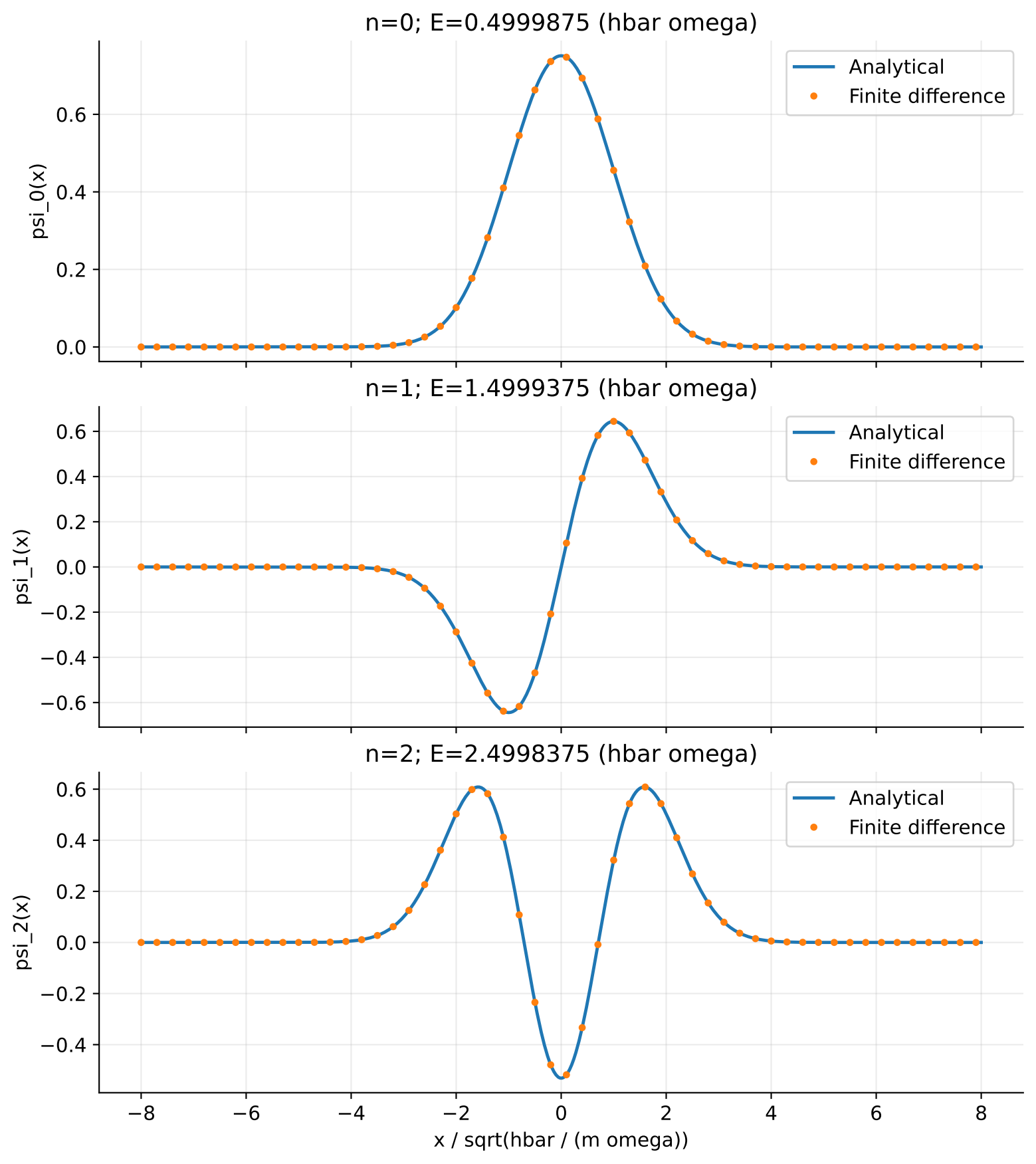
n=0; E=0.4999875 (hbar omega)
Analytical
Finite difference
psi_0(x)
0.0
0.2
0.4
0.6
n=1; E=1.4999375 (hbar omega)
Analytical
Finite difference
psi_1(x)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
n=2; E=2.4998375 (hbar omega)
Analytical
Finite difference
psi_2(x)
0.6
0.4
0.2
0.0
−0.2
−0.4
−8
−6
−4
−2
0
2
4
6
8
x / sqrt(hbar / (m omega))